

Small-molecule permeation and condensate-membrane interactions probed by molecular simulations

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The efficacy of many biomolecular therapeutics, such as peptides and mRNAs, rely on their efficient intracellular delivery, which is frequently hindered by the poor membrane permeability and/or low endosomal escape efficiency of macromolecules. Recent studies on condensates that arise from liquid-liquid phase separation (LLPS) indicate that condensates formed by phase-separating peptides or fluorescent molecules can recruit macromolecular cargoes and deliver them intracellularly with high efficiency. However, our understanding on the mechanism by which the cargo-carrying condensates enter the cell and how condensate-membrane interactions can be tuned by physico-chemical properties of the phase-separating molecules remain limited. In this talk, I will first discuss free energy calculations aimed at elucidating the permeation energetics of small molecules and then present our recent simulations of a photo-responsive, phase-separating fluorescent molecule (PPFM) based on pyrene that undergoes LLPS in the aqueous solution and recruit payload molecules into the resulting condensate. Through simulations of the condensate-membrane systems and drawing analogy to small-molecule permeation, I will discuss how PPFMs may achieve high loading efficiency and enable the co-transport of their payload molecules across lipid membranes.